

Lab 2 – Dataset and Problem Summary

1.Dataset

Name: Student Performance & Study Habits Dataset
Source: Kaggle
Format: CSV, 1,000 student records

The dataset contains 1,000 student records with 12 columns:

- student_id — unique identifier
- gender
- study_time_hours
- attendance_percent
- sleep_hours
- parental_education (has some missing values)
- internet_access
- extracurricular_activities
- part_time_job
- previous_grade
- final_exam_score — the target variable for this lab
- final_grade — a letter-grade version of the outcome (not used here)

2.Machine Learning Problem

Problem Type	Regression problem.
Target Variable	final_exam_score which is a continuous numeric value
Features	gender, study_time_hours, attendance_percent, sleep_hours, parental_education, internet_access, extracurricular_activities, part_time_job, previous_grade
Goal	The model is expected to learn the relationship between a student's study habits, lifestyle, and background, and use that relationship to predict their final exam score as a number.

3. Methodolgy Steps

Methodology: Predicting Final Exam Score (Regression)

